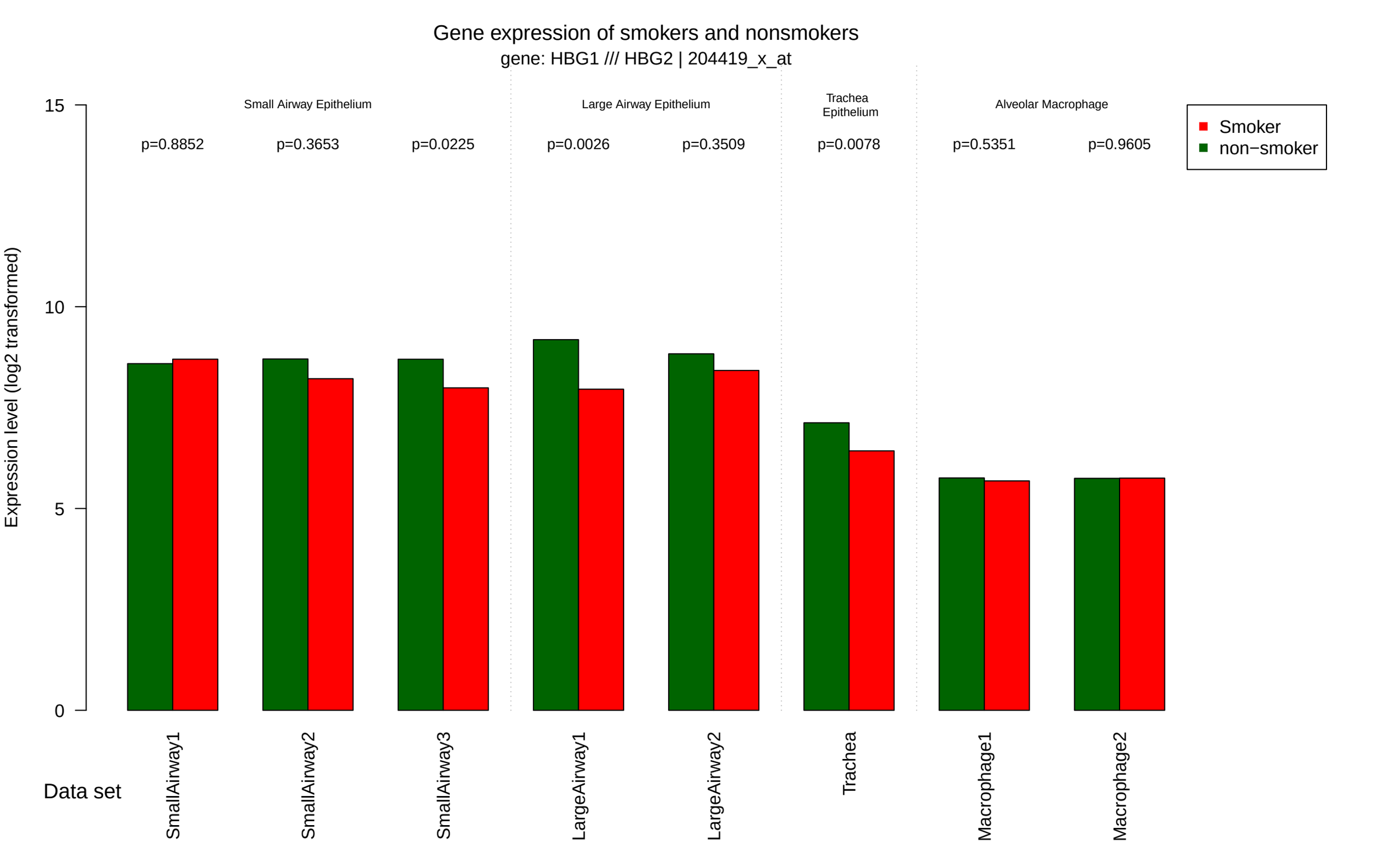
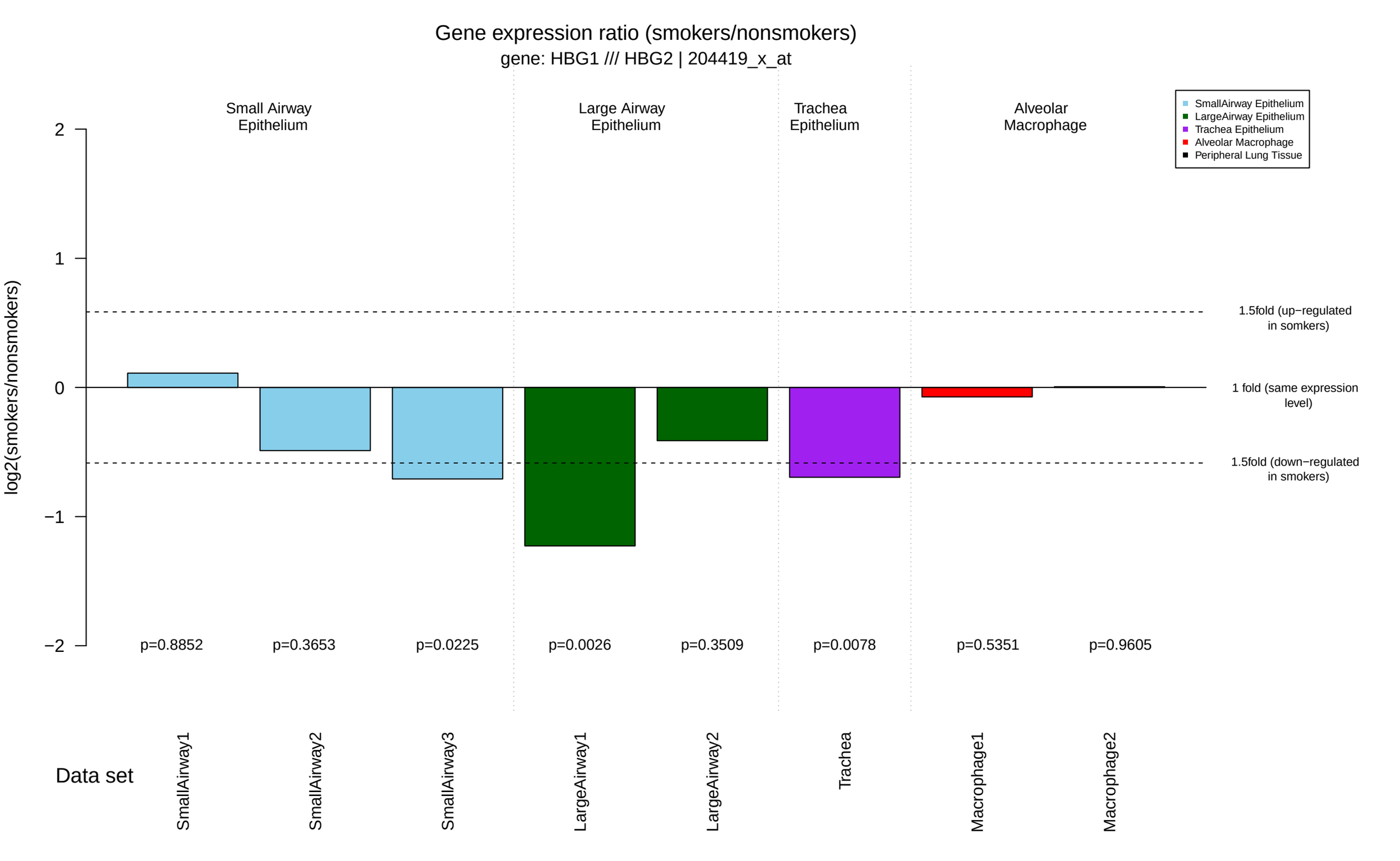
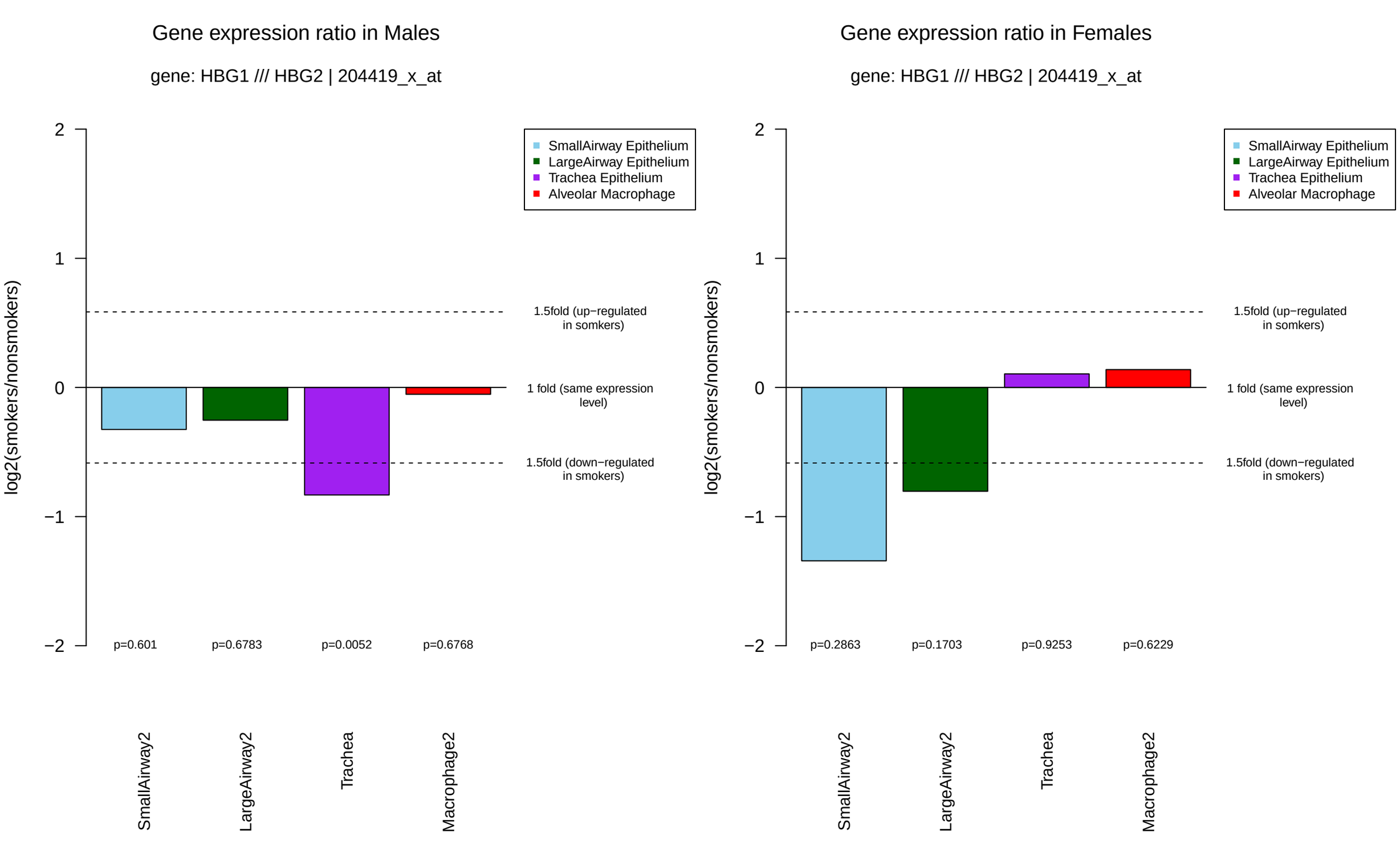
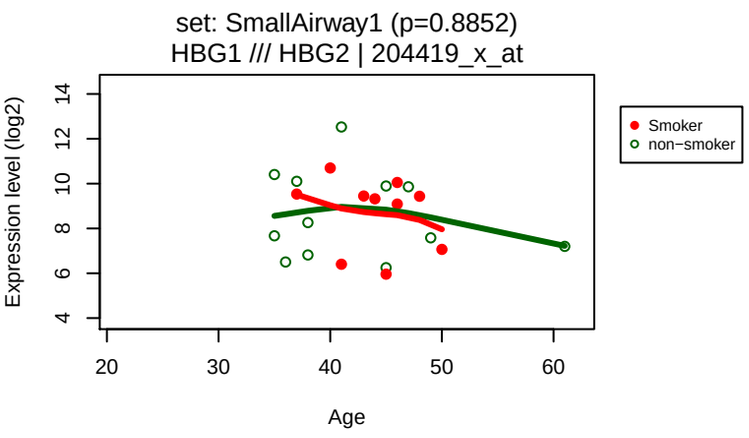
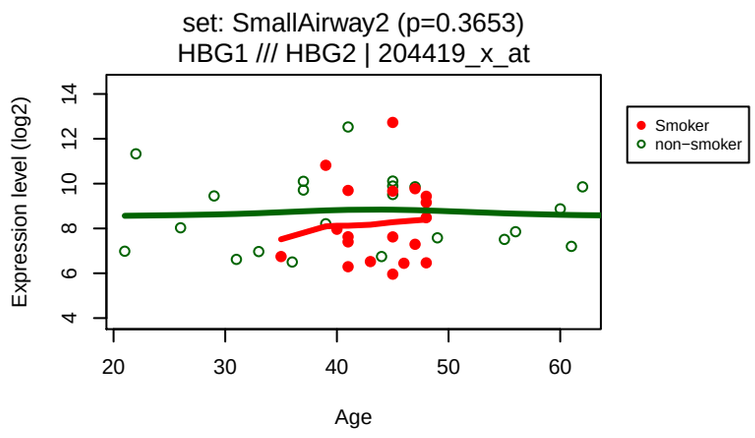
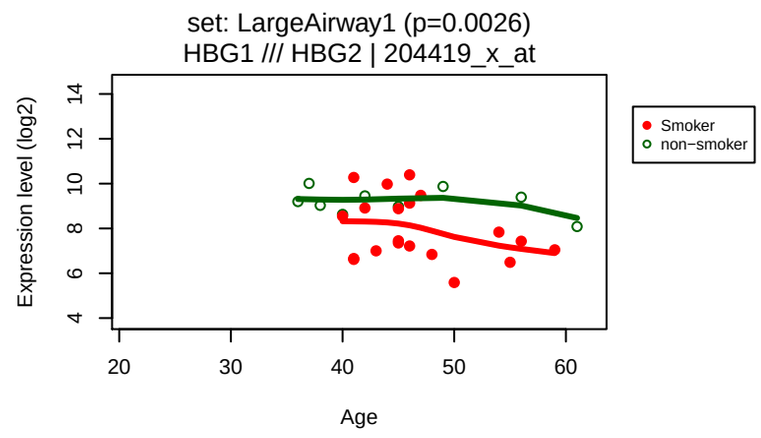

Gene expression of smokers and nonsmokers
gene: HBG1 /// HBG2 | 204419_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8852
p=0.3653
p=0.0225
p=0.0026
p=0.3509
p=0.0078
p=0.5351
p=0.9605
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

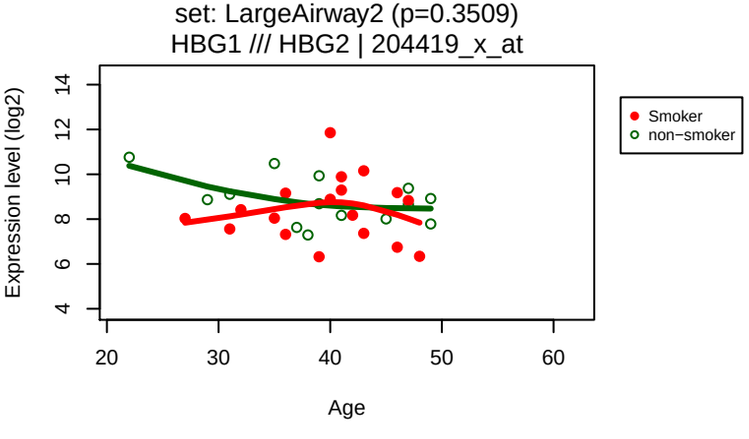

Gene expression ratio (smokers/nonsmokers)
gene: HBG1 /// HBG2 | 204419_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8852
p=0.3653
p=0.0225
p=0.0026
p=0.3509
p=0.0078
p=0.5351
p=0.9605
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

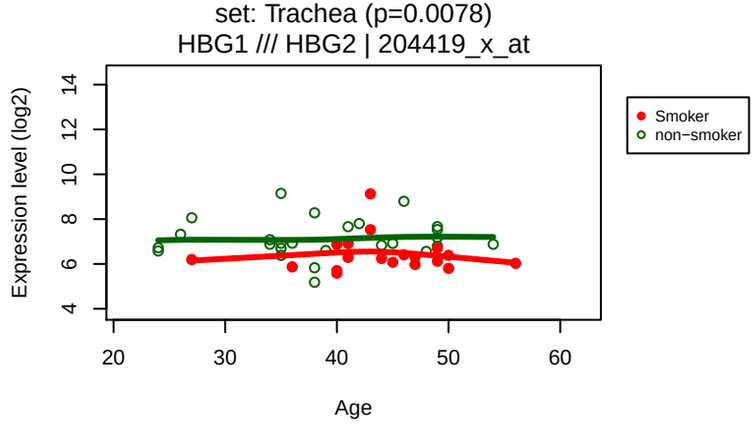

Gene expression ratio in Males
gene: HBG1 /// HBG2 | 204419_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.601
p=0.6783
p=0.0052
p=0.6768
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HBG1 /// HBG2 | 204419_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2863
p=0.1703
p=0.9253
p=0.6229
SmallAirway2
LargeAirway2
Trachea
Macrophage2

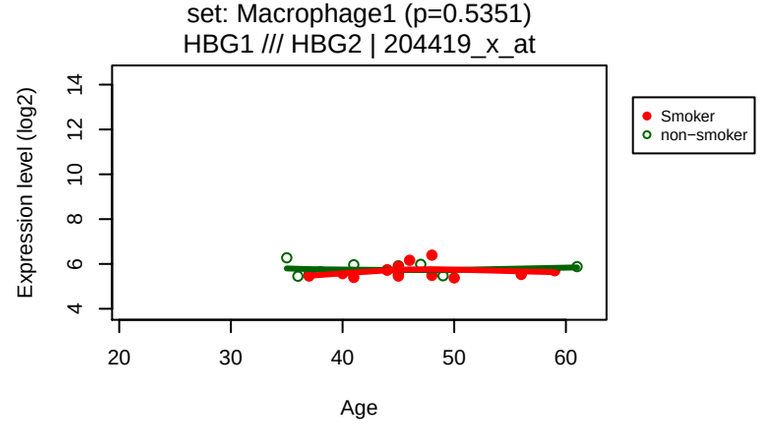

set: SmallAirway1 (p=0.8852)
HBG1 /// HBG2 | 204419_x_at
Expression level (log2)
Age
Smoker
non−smoker

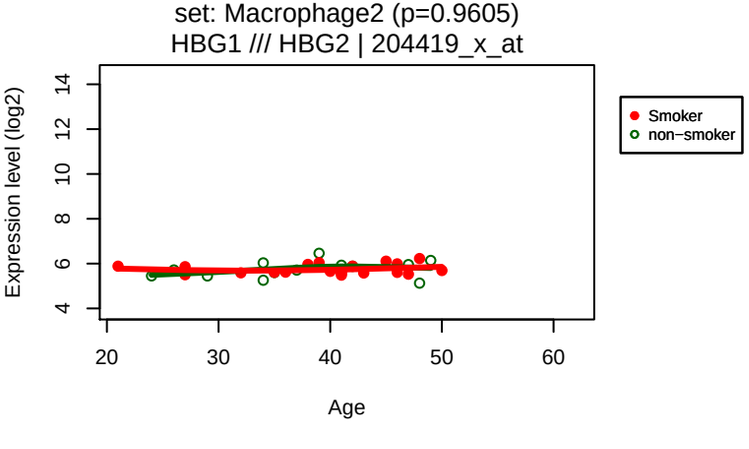

set: SmallAirway2 (p=0.3653)
HBG1 /// HBG2 | 204419_x_at
Expression level (log2)
Age
Smoker
non−smoker

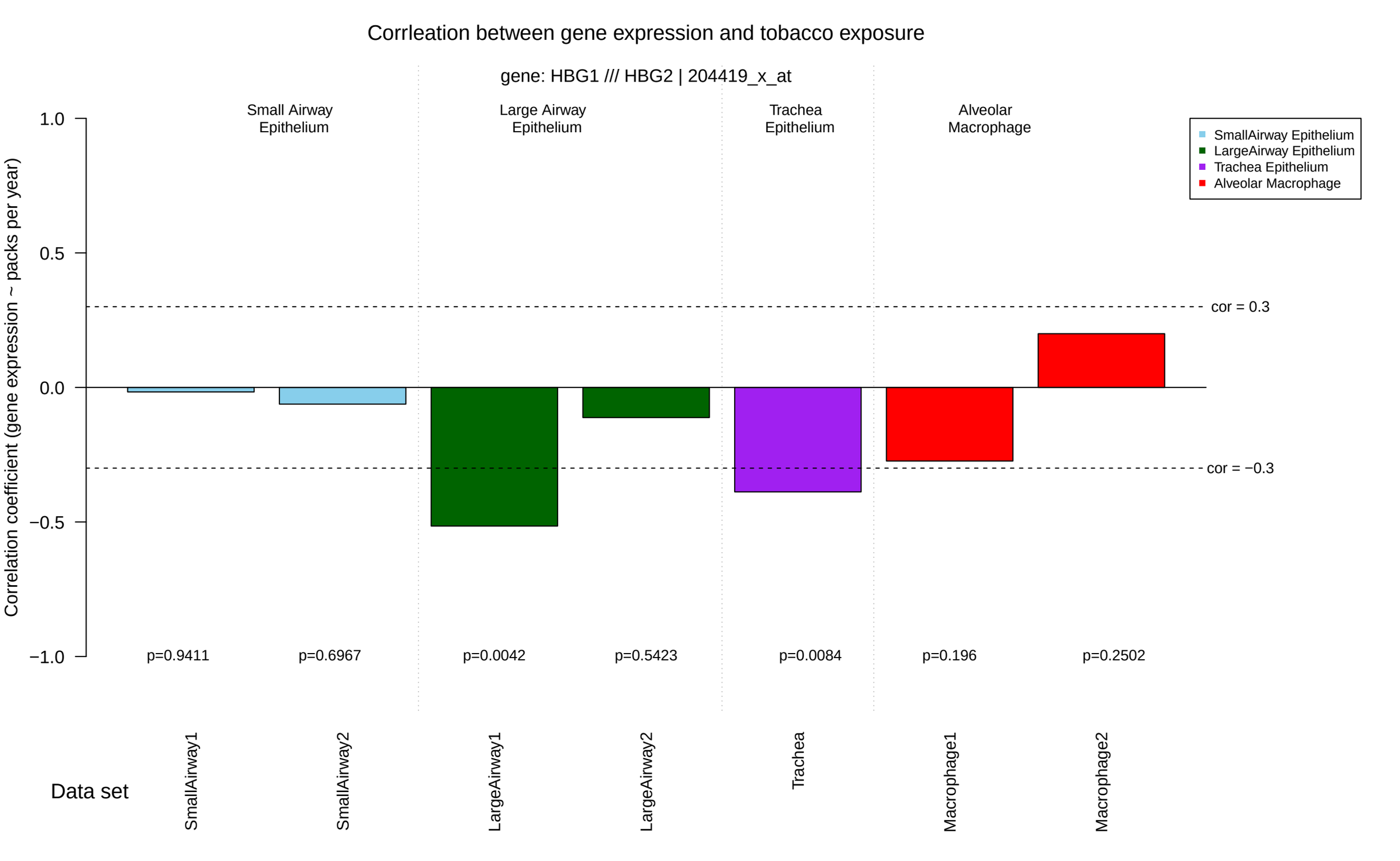

set: LargeAirway1 (p=0.0026)
HBG1 /// HBG2 | 204419_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3509)
HBG1 /// HBG2 | 204419_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0078)
HBG1 /// HBG2 | 204419_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5351)
HBG1 /// HBG2 | 204419_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9605)
HBG1 /// HBG2 | 204419_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HBG1 /// HBG2 | 204419_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9411
p=0.6967
p=0.0042
p=0.5423
p=0.0084
p=0.196
p=0.2502
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2